The choice of a compiler is justified when the programmer is confident that the program is as near error-free as possible because:

- an executable file can be created
- this can be distributed for general use
- execution of the program will be faster than if an interpreter were used.

Java

When the programming language Java was created, a different philosophy was applied to how it should be used. Each different type of computer has to have a Java Virtual Machine created for it. Then when a programmer writes a Java program this is compiled first of all to create what is called Java Byte Code. When the program is run, this code is interpreted by the Java Virtual Machine. The Java Byte Code can be transferred to any computer that has a Java Virtual Machine installed.

8.06 Features found in a typical Integrated Development Environment (IDE)

Whatever language is used for writing source code and whatever compiler or interpreter is being used there will be one or more IDEs available to assist the programmer. This section discusses the types of feature that should be provided by an IDE.

Prettyprinting

Prettyprint refers to the presentation of the program code typed into an editor. For example, the Python IDLE (see Figure 8.02) automatically colour-codes keywords, built-in function calls, comments, strings and the identifier in a function header. In addition, indentation is automatic.

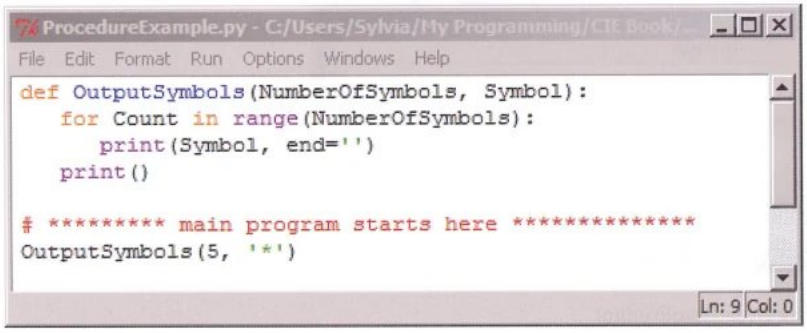


Figure 8.02 Prettyprint in the Python IDLE

Context-sensitive prompts

This feature displays hints (or a choice of keywords) and available identifiers that might be appropriate at the current insertion point of the program code. Figure 8.03 shows an example of the Visual Studio editor responding to text typed in by the programmer.

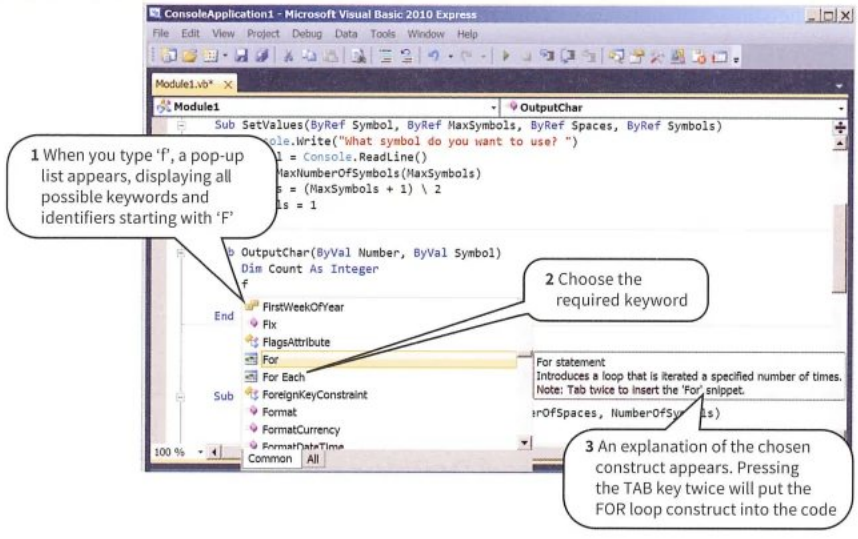


Figure 8.03 Context-sensitive prompts in the Visual Studio editor

Dynamic syntax checks

When a line has been typed, some editors perform syntax checks and alert the programmer to errors.

Figure 8.04 shows an example of the Visual Studio editor responding to a syntax error.

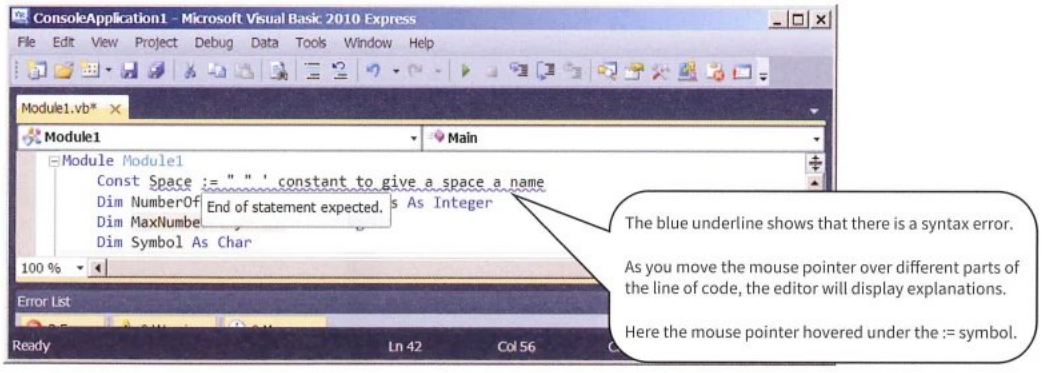


Figure 8.04 Dynamic syntax check in the Visual Studio editor

Expanding and collapsing code blocks

When working on program code consisting of many lines of code, it saves excessive scrolling if you can collapse blocks of statements.

Debugging

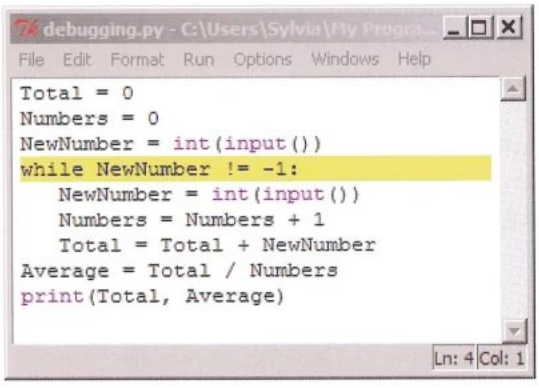
An IDE often contains features to help with **debugging**.

KEY TERM

Debugging: finding and correcting errors, often called 'bugs', in a program

If a Debugger feature has been switched on it is possible to select a breakpoint. When the program starts running it will stop when it reaches the breakpoint. The program can then be stepped through, one instruction at a time. Figure 8.05 shows the windows presented to the user in the Python IDLE when this feature is being used.

a



b

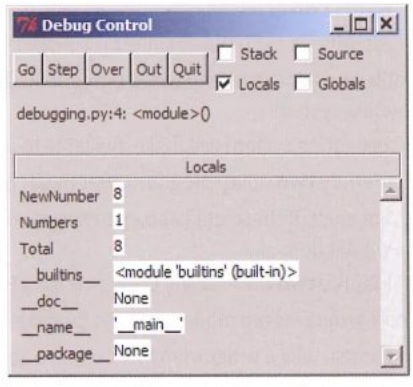


Figure 8.05 (a) A Python program showing a breakpoint; (b) the Debug Control window

TASK 8.03

Investigate the facilities in the editors you have available. If you have a choice of editors, you may like to use the editor with the most helpful facilities.

Reflection Point:

Much of the discussion in this chapter only summarises what an operating system does. Do you think it would be helpful to move on immediately to have a look at some of the content in sections 20.01 to 20.05 of Chapter 20?

Summary

- Operating system tasks can be categorised in more than one way, for example, some are for helping the user, others are for running the system.
- Utility programs include hard disk utilities, backup programs, virus checkers and file compression utilities.
- Library programs, including Dynamic Link Library (DLL) files, are available to be incorporated into programs; they are usually subroutines and are very reliable.
- A high-level language can be translated using an interpreter or a compiler.
- A Java compiler produces Java Byte Code which is interpreted by a Java Virtual Machine.
- An integrated Development Environment (IDE) contains many features that provide support for a programmer when a program is being written and when it is being corrected.

Exam-style Questions

1 **a** One of the reasons for having an operating system is to provide a user interface to a computer system.

i Name **two** different types of interface that an operating system should provide. [2]

ii Identify for each type of interface a device that could be used to enter data. [2]

b Identify and explain briefly **three** other management tasks carried out by an operating system. [6]

2 **a** A PC operating system will make available to a user a number of utility programs.

i Identify **two** utility programs that might be used to deal with a hard disk problem. [2]

ii For each of these utility programs explain why it might be needed and explain what it does. [5]

iii Identify **two** other utility programs for a PC user. [2]

b Library programs are made available for programmers.

i Explain why a programmer should use library programs. [3]

ii Identify **two** examples of a library program. [2]

3 Assemblers, compilers and interpreters are examples of translation programs.

a State the difference between an assembler and a compiler or interpreter. [1]

b A programmer can choose to use an interpreter or a compiler.

i State **three** differences between how an interpreter works and how a compiler works. [3]

ii Discuss the advantages and disadvantages of an interpreter compared to a compiler. [4]

iii If a programmer chooses Java, a special approach is used. Identify **one** feature of this special approach. [1]

4 **a** Explain the meaning of the following terms:

Prettyprinting

Context-sensitive prompt

Dynamic syntax check

Debugging [8]

b Describe the features you would expect a debugger to provide. [4]

5 Before it is used, a hard disk is formatted using disk formatter software.

a Explain why formatting is needed. [2]

b Eventually, the performance of the hard disk deteriorates.

Name **three** other utility programs that might be required. State why each is needed. [6]

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6 A programmer is writing a program that includes code from a program library.

a Describe **two** benefits to the programmer of using one or more library routines. [4]

b The programmer decides to use a Dynamic Link Library (DLL) file.

i Describe **two** benefits of using DLL files. [4]

ii State **one** drawback of using DLL files. [2]

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